

Logic Dungeon II

THE SYLLOGISM LABYRINTH

Purpose: Use the logic tools from Weeks 4–6 to select the best-supported route through the labyrinth.

The Second Seal

Beyond the Hall of Inscriptions, the expedition reaches a circular chamber. The next seal lies in the **Upper Archive**. Three routes leave the chamber:

- **Raven Stair** — an upward arrow is cut beneath a raven seal.
- **Well Passage** — cold air flows through a round stone opening.
- **Lantern Gallery** — a blue survey thread disappears around its first turn.

Each route has an argument in its favor. A valid form may rest on a bad premise. A true conclusion may be supported by an invalid argument. The expedition must test both the bridge and the ground beneath it.

Two-test rule: First apply the *form test*: if the premises were granted, would the conclusion follow? Then apply the *fact test*: are the premises true, verified, doubtful, or false? The best route needs the strongest combination of form and facts.

Tools available: premises and conclusion; major, minor, and middle terms; practical syllogism map; A/E/I/O mood; figure; undistributed middle; two negative premises; validity; truth; soundness; premise pressure point.

A perfect bridge drawn on rotten paper will not carry an expedition.

Part 1: The Raven Inscription

All routes bearing an Archive seal are routes that reach the Upper Archive.

The Raven Stair is a route bearing an Archive seal.

Therefore, the Raven Stair is a route that reaches the Upper Archive.

Part 1 Q1: List the major premise, minor premise, and conclusion. Name the major, minor, and middle terms.

Part 1 Q1 ANSWER BOX

Part 1 Q2: Explain how the middle term is supposed to connect Raven to the Upper Archive. What premise becomes the pressure point for the route decision, and what evidence would test it?

Part 1 Q2 ANSWER BOX

Part 2: The Breath of the Well — Danger Station

A guard places a hand in the cold draft and argues:

All passages opening from the Upper Archive carry cold air.
The Well Passage carries cold air.
Therefore, the Well Passage opens from the Upper Archive.

Part 2 Q1: Identify the mood and the placement of the middle term. Is the form valid? Explain whether the middle term connects the route to the Archive.

Part 2 Q1 ANSWER BOX

Part 2 Q2: What may the expedition record about Well after the form test? Preserve any true observation, but state what the argument has not proved.

Part 2 Q2 ANSWER BOX

Part 3: The Blue-Thread Rule

A polished brass plaque beside Lantern reads:

All routes marked by current blue survey thread are routes that lead to the Upper Archive.
The Lantern Gallery is marked by current blue survey thread.
Therefore, the Lantern Gallery leads to the Upper Archive.

Part 3 Q1: Identify the mood and figure pattern using M, P, and S. Apply the form test.

Part 3 Q1 ANSWER BOX

Part 3 Q2: Apply the fact test using only what is known so far. Classify each premise as established or doubtful, identify the pressure point, and give the argument's present diagnosis: sound, unsound, or not yet proved.

Part 3 Q2 ANSWER BOX

Part 4: The Survey Cabinet

The expedition opens a locked cabinet bearing today's date. Inside are three records:

Record A — Chief Surveyor's Rule: "Every route carrying an unbroken blue thread from this cabinet was inspected today and terminates in the Upper Archive." The seal and signature match the cabinet log.

Record B — Thread Check: "Blue thread spool 3 runs unbroken from this cabinet into Lantern Gallery." Two surveyors signed the check.

Record C — Mason's Note: "Raven seals mark routes proposed for Archive access during the old construction. Some were completed; some were abandoned."

Part 4 Q1: Use Records A and B to rebuild the Lantern syllogism in standard order. Identify its mood, middle term, and whether it now passes both the form and fact tests.

Part 4 Q1 ANSWER BOX

Part 4 Q2: How does Record C affect the Raven argument? State whether the form changes, whether the pressure-point premise is established, and what the expedition may now record about Raven.

Part 4 Q2 ANSWER BOX

Part 5: Two Tempting Shortcuts

Shortcut A: No flooded passages are dry passages.
No Lantern routes are flooded passages.
Therefore, all Lantern routes are dry passages.

Part 5 Q1: State the mood. Is the form valid? Explain what two negative premises fail to connect. Does this argument add dependable support for Lantern?

Part 5 Q1 ANSWER BOX

Shortcut B: All routes to the Upper Archive rise.
The Lantern Gallery rises.
Therefore, the Lantern Gallery leads to the Upper Archive.

Part 5 Q2: Explain why this is weak proof even if Lantern really does reach the Archive. Identify the form error and distinguish a true conclusion from a good argument.

Part 5 Q2 ANSWER BOX

Part 6: The Cistern Report

A maintenance report states: “The Well Passage crosses above an ice cistern. The cistern sends cold air into several passages on both upper and lower levels.” The report is current and signed.

Part 6 Q1: Apply the fact test to the Well argument. Is the cold-air premise true, false, or doubtful? Does the new fact repair the invalid form? Explain what the report contributes to the route decision.

Part 6 Q1 ANSWER BOX

Part 6 Q2: Compare these diagnoses: *valid but doubtful*, *invalid with a true observation*, and *sound*. Assign each diagnosis to Raven, Well, or Lantern using the strongest evidence now available.

Part 6 Q2 ANSWER BOX

Part 7: Labyrinth Evidence Chart

Route	Form	Pressure point	Fact status	Final diagnosis
Raven				
Well				
Lantern				

Part 7 Q1: Beneath the chart, identify one valid argument with a weak premise, one invalid argument containing a true fact, and the strongest sound argument. Explain why these are different.

Part 7 Q1 ANSWER BOX

Part 8: The Route Decision

Part 8 Q1: Which route is best supported as the way to the Upper Archive? Use at least three pieces of evidence from different parts. Explain why each rejected route has weaker proof.

Part 8 Q1 ANSWER BOX

Part 8 Q2: Name the next observation that would most improve confidence before the expedition commits all twenty adventurers. Then explain what logic has established and what it has not guaranteed.

Part 8 Q2 ANSWER BOX

Part 9: The Last Voice in the Labyrinth

As the company prepares to enter Lantern, the stone guide says:

“The Lantern argument is valid. Therefore every premise on the plaque must be true, and the route must be safe.”

Part 9 Q1: Separate the guide’s claims. Explain why validity does not establish premise truth or route safety. Repair the statement into a conclusion the evidence supports.

Part 9 Q1 ANSWER BOX

Part 9 Q2: In 2–3 sentences, explain why the expedition needed both the form test and the fact test to escape the Syllogism Labyrinth.

Part 9 Q2 ANSWER BOX

The blue thread passes through the chosen arch. The second seal waits above.